AssistVision: A Turntable–Load-Cell Rig and a Controlled Benchmark of Monocular versus Multi-View Food Volume Estimation

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***Abstract*—Image-based portion estimation needs metric volume, and a single image does not contain metric scale. We build a low-cost capture rig ($\leq$ INR 4,220 at indicative prices; ESP32-S3 camera, stepper turntable, HX711 load cell, printed ArUco mat) that returns N calibrated views together with the true mass of the plate, and we ask, with controlled experiments rather than anecdotes, when a single-view depth network can replace multi-view reconstruction. On the 507-dish Nutrition5k depth test split we benchmark three monocular depth models under three explicitly separated calibration protocols (raw, deployable anchor, oracle upper bound) and report bootstrap confidence intervals and paired tests. On 11 synthetic plates with exact mesh volume we compare dense multi-view stereo against exact volume and specify ablations of view count, sensor resolution, JPEG quality, marker-scale error and plate material. *These are partial-run results: three of eleven planned depth models and 11 of 22 synthetic scenes had finished at the time of writing.* Deployable table-only anchoring of relative models fails from an overhead view (median volume error $\geq$94.8 %), because a fronto-parallel plane cannot fix both scale and shift. With an oracle fit the best model (DA-V2-S) reproduces sensor volume to 14.1 % median error when the sensor's support plane is used, but to only 37.4 % when the plane is fitted in its own prediction. Dense 24-view MVS with marker scale reaches 3.9 % [1.0, 5.7] median absolute error against exact mesh volume. Even sensor-depth volume predicts weighed mass worse than the dataset mean (68.2 vs 52.0 % median error), which is why the rig weighs the plate. No result in this paper comes from the physical rig.**

I. Introduction

Dietary assessment by photograph is attractive because it removes manual logging, but portion size—grams—is the quantity nutrition depends on and the one images recover worst. Perspective projection discards absolute scale: a small plate near the camera and a large plate far away produce the same pixels. Any system must inject metric scale from outside the image: from a reference object, from a depth sensor, from camera motion with known geometry, or from a learned prior. This paper measures how much each of these choices costs.

Our contributions are:

1) An open hardware capture rig (Sec. IV) that produces multi-view images, metric scale and gravimetric ground truth in one button press, with firmware, host software, wiring and a bench-test protocol.
2) A benchmark of monocular depth models for food on real RGB-D data (Nutrition5k) that separates deployable calibration from oracle calibration, a distinction prior food-volume work often blurs.
3) A controlled synthetic study with exact mesh volumes comparing dense MVS against ground truth, plus an ablation protocol for the factors the rig designer controls: views, resolution, compression, marker accuracy and serveware material.
4) A documented record of design decisions and rejected alternatives (Sec. III), each tied to a measurement or a stated assumption.

II. Related Work

Food volume. Nutrition5k [1] provides overhead RGB-D video and weighed dishes; its depth-based volume baseline uses a constant per-pixel footprint at the capture-plane distance. Multi-view food reconstruction has used structure-from-motion with reference cards and dense stereo [2], [3]. **Monocular depth.** Relative depth models (MiDaS [4], Depth Anything [5], [6]) predict depth up to an unknown affine transform of inverse depth; metric models (ZoeDepth [7], Depth Pro [8], metric-finetuned Depth Anything V2) predict absolute depth from priors learned on room- and street-scale scenes. **Reconstruction.** We use COLMAP [9] for SfM and OpenMVS for dense MVS.

III. Design Decisions and Rejected Alternatives

Table I summarises each decision; the evidence column points to the experiment that supports it. Decisions marked *assumption* rest on datasheets or cost quotes, not on our measurements.

IV. Hardware

The rig (Figs. 1–3) uses an ESP32-S3 camera board (OV2640, up to 1600 × 1200), a 28BYJ-48 stepper through a ULN2003 driver (4076 half-steps per revolution, 0.088° per step), and a 5 kg bar load cell read by an HX711 at 10 Hz. The firmware exposes `/capture`, `/rotate`, `/weight`, `/tare`, `/calibrate` and camera-setting endpoints. Exposure, gain and white balance are locked after convergence so that all

TABLE I
DESIGN DECISIONS, ALTERNATIVES CONSIDERED, AND THE EVIDENCE BEHIND EACH CHOICE.

Decision	Chosen	Rejected alternatives	Why / evidence
Scale source	Printed ArUco mat, multi-view triangulation	Credit-card reference in one image; metric monocular depth; depth sensor	Volume scales with the *cube* of length error (Sec. V-B); metric networks are off by large factors at plate distance (Table II); a D435-class sensor costs $\sim$10$\times$ the camera (assumption).
Geometry	Dense MVS (COLMAP + OpenMVS)	Sparse SfM points only; single-view depth	Sparse points under-cover low-texture food (Sec. VI-C); single view cannot see the far side of a mound even with perfect depth.
Capture motion	Fixed camera, motorised turntable	Hand-held phone orbit; multi-camera ring	Repeatable baselines and known angles; one camera keeps cost and synchronisation trivial; the angle prior is checked against SfM.
Camera tilt	45° elevation	Straight overhead (Nutrition5k)	Overhead view makes the support plane fronto-parallel: the relative-depth anchor fit becomes rank-deficient (condition number 10^{14}, Sec. V-C).
Controller	ESP32-S3 + OV2640	Raspberry Pi + camera; smartphone app	Pi adds cost, boot time and an OS to maintain; phones vary per user and cannot drive the turntable or scale. Processing is host-side, so the MCU only needs camera, Wi-Fi and GPIO (assumption: cost table in hardware README).
Ground truth	5 kg bar load cell + HX711 under the plate	External kitchen scale; no mass at all	In-rig weighing removes a manual step and lets every scan carry its own reference; tare-before-serve gives food-only mass.
Volume integral	Height field over the fitted support plane	Constant per-pixel footprint (Nutrition5k); voxel carving; mesh + watertight closure	Exact on tilted planes in our unit tests (0.09 % error) where the constant-footprint formula is not; shared across all methods so comparisons measure geometry, not bookkeeping.
Mono calibration	Report raw / anchor / oracle separately	Report only oracle-aligned numbers	Oracle alignment uses the ground truth being evaluated; it is an upper bound, not a deployable result.

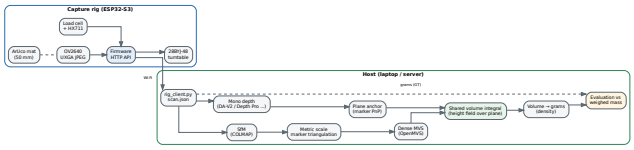


Fig. 1. System architecture. The rig exposes an HTTP API; the host drives the scan loop, so dropped frames are retried rather than corrupting a scan.

views of a scan share photometry, which MVS assumes. Stepper coils are released while weighing to avoid heating and vibration of the load cell. Tare and the counts-per-gram calibration persist in flash. The firmware compiles for the ESP32-S3 target with the Arduino core 3.3.12; a fake-firmware test exercises every client endpoint, retry on a dropped frame, and calibration arithmetic.

Physical validation (pending). Load-cell linearity, repeatability and drift, turntable angular error, scan duration and transfer latency are specified as scripted bench tests (`rig_client.py bench-*`) whose results are not yet measured, because the rig has not yet been assembled. We report no physical number we have not measured.

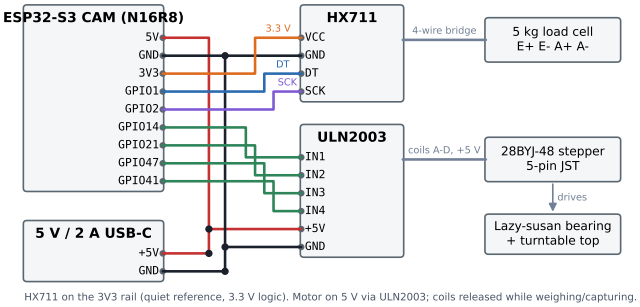


Fig. 2. Wiring of the ESP32-S3, HX711 load-cell amplifier and ULN2003 stepper driver.

V. METHOD

A. Shared volume integral

All methods produce a 3-D point set P in metres and a support plane $(\mathbf{n}, d)$ oriented towards the camera. Each point gets a height $h = \mathbf{n}^\top \mathbf{p} + d$ and plane coordinates in an orthonormal basis; points are binned into square cells of side c and each cell keeps the median height. Empty cells inside

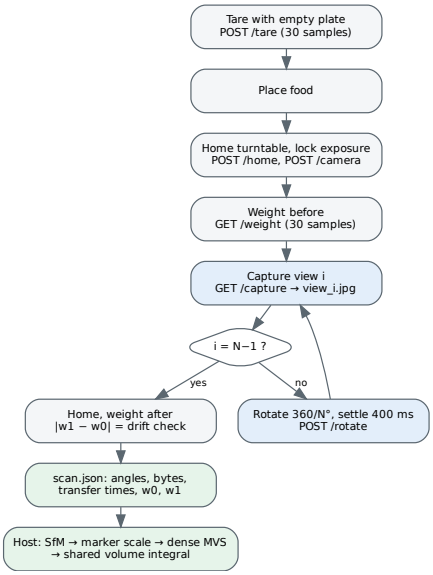


Fig. 3. Scan protocol implemented by `rig_client.py scan`. Weight is read before and after the turntable moves; their difference is a per-scan drift/vibration check.

the footprint are filled by nearest neighbour, negative heights are clipped, and

$$V = c^2 \sum_{\text{cells}} \max(h, 0). \tag{1}$$

Because monocular, multi-view and sensor depth all share this integral, differences in volume reflect differences in geometry, not in post-processing.

B. Metric scale from the marker mat

Marker corners are detected in every view and triangulated with the SfM cameras; the scale s is the ratio of the known side to the median triangulated side. A relative error ϵ in s produces a volume error of $(1+\epsilon)^3 - 1 \approx 3\epsilon$, so a 1 % scale error costs about 3 % in volume; Sec. VI-C measures this directly.

C. Monocular protocols

Raw: the model's metric output is used as-is. **Anchor**: the prediction is calibrated using only the support plane (the table in Nutrition5k, the marker mat in our rig), never the food; metric outputs get one scale, relative outputs an affine map of inverse depth. **Oracle**: the same maps fitted to ground-truth depth over the whole dish region—an upper bound that is not available at deployment. For relative models we record the condition number of the anchor fit; a fronto-parallel plane has nearly constant inverse depth, so scale and shift are not separately identifiable.

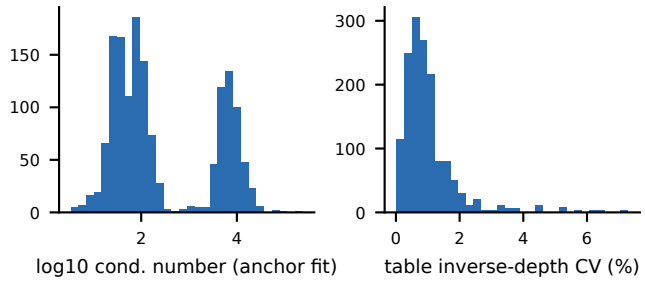


Fig. 4. Anchor-fit conditioning on Nutrition5k (relative models): condition number of the affine fit (left) and coefficient of variation of table inverse depth (right). A low CV means the table alone cannot separate scale from shift.

VI. Experiments

A. E1: Monocular depth on Nutrition5k

Anchoring on the table fails from overhead. All three relative models have a median volume error of 94.8–99.4 % under the deployable anchor protocol (Table II), even though their predicted table is flat to 0.6–0.7 mm RMS: the affine fit reproduces the table and collapses the food relief. The anchor fit is badly conditioned on most dishes (Fig. 4), as expected when the only reference is a plane almost perpendicular to the optical axis. This is the quantitative reason the rig views the plate at 45° and uses the marker mat and multiple views.

Depth metrics do not predict volume. AbsRel is 0.016–0.045 for every model and protocol, yet volume errors span 14–99 %. Food relief (1–3 cm) is a small fraction of the 36 cm camera distance, so a model can score well on depth and still lose most of the volume. We therefore rank models on volume, not on AbsRel.

Oracle upper bound. With an oracle fit DA-V2-S is best: 37.4 % [33.0, 41.9] median volume error with its own support plane and 14.1 % with the sensor's plane. The paired Wilcoxon test with Holm correction separates it from DA-V1-S ($p = 5.2 \times 10^{-11}$) and MiDaS-DPT-Hybrid ($p = 5.5 \times 10^{-53}$). Under the anchor protocol DA-V1-S and DA-V2-S are not distinguishable ($p = 0.31$). The gap between the two plane choices shows that fitting the support plane in predicted depth causes more than half of the error; in the rig the marker mat provides this plane independently of the network.

Coverage and failures. 498 of the 503 usable dishes were evaluated for every model. The remaining five failed in the plane fit (degenerate table region) and are counted as failures, not dropped. Latency is 1.4–1.9 s per image on the VM CPU, measured while other jobs shared the machine (VM evidence, an upper bound). The metric models (ZoeDepth, metric DA-V2, Depth Pro) are implemented in the same benchmark but had not finished at the time of writing, so the raw protocol is not yet reported.

B. E2: Sensor-referenced volume and weighed mass

Nutrition5k has no food-volume label, so the volume target in E1/E2 is the volume integrated from RealSense depth over the dish region with the same height-field integral (Eq. 1)

TABLE II

E1 (NUTRITION5K, n DISHES PER MODEL AS SHOWN). DEPTH ERROR OVER THE DISH REGION AND VOLUME ERROR RELATIVE TO SENSOR-DEPTH VOLUME, PER MODEL AND CALIBRATION PROTOCOL. BRACKETS: 95% BOOTSTRAP CI (10 000 RESAMPLES). LATENCY IS CPU TIME PER IMAGE ON THE DEVELOPMENT VM.

Model	Prot.	AbsRel	h MAE (mm)	Vol. APE med. (%)	$\leq$20% (%)	APE, sensor plane (%)	table RMS (mm)	s/img
DA-V1-S	anchor	0.039	13.6	94.8 [92.9, 96.0]	3	86.1	0.6	1.64
DA-V2-S	anchor	0.038	13.4	96.0 [93.4, 97.8]	3	86.9	0.6	1.85
MiDaS-DPT-Hybrid	anchor	0.045	15.8	99.4 [98.9, 99.8]	1	93.7	0.7	1.39
DA-V2-S	oracle	0.016	12.0	37.4 [33.0, 41.9]	26	14.1	2.6	1.85
DA-V1-S	oracle	0.017	10.3	47.6 [42.9, 51.8]	17	17.2	2.4	1.64
MiDaS-DPT-Hybrid	oracle	0.022	11.4	86.9 [82.7, 89.0]	3	33.8	1.1	1.39

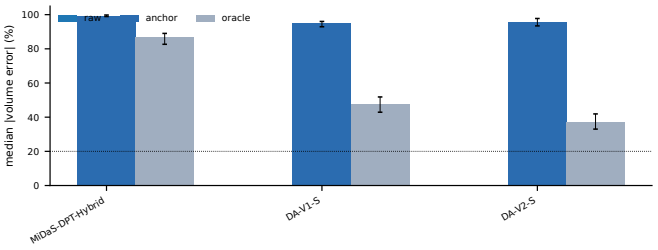


Fig. 5. Per-dish absolute volume error versus sensor depth, by model and protocol (Nutrition5k).

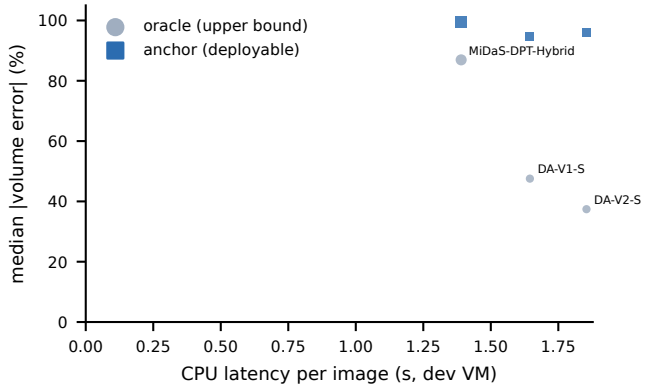


Fig. 6. Accuracy–latency trade-off (anchor protocol; CPU latency on the development VM).

applied to each model's prediction. The error therefore isolates the depth model: a model with zero error here reproduces the sensor, whatever the sensor's own error on food is. Even sensor-depth volume with a cross-validated density predicts mass worse than ignoring the image and predicting the mean (Table III). Sensor volume and weighed mass are only weakly correlated (Pearson $r = 0.19$, Spearman $\rho = 0.17$, $n = 503$). Mixed dishes differ in density, and the dish region includes the plate rim. A single image-derived volume therefore cannot stand in for grams on real dishes, and this is the quantitative case for weighing the plate in the rig. Mass counts n vary because dishes with a non-positive predicted volume are excluded from the density fit.

TABLE III

E2 (NUTRITION5K). MASS PREDICTED AS ρV WITH A SINGLE DENSITY ρ FITTED ON FOUR FOLDS AND SCORED ON THE HELD-OUT FOLD (5-FOLD CV). BRACKETS: 95 % BOOTSTRAP CI.

Volume source	n	MAE (g)	median APE (%)
RealSense depth (sensor ceiling)	503	139.3 [129.3, 150.0]	68.2 [63.1, 72.9]
DA-V2-S, oracle	483	178.1 [161.6, 196.6]	85.7 [84.2, 87.5]
DA-V1-S, oracle	491	180.4 [167.2, 194.1]	88.2 [85.7, 89.6]
MiDaS-DPT-Hybrid, oracle	490	193.2 [178.7, 208.5]	96.9 [95.9, 97.6]
DA-V2-S, anchor	490	189.0 [175.8, 202.7]	98.7 [97.9, 99.3]
DA-V1-S, anchor	487	201.6 [186.4, 218.5]	99.5 [99.4, 99.7]
MiDaS-DPT-Hybrid, anchor	441	194.0 [180.1, 208.4]	99.9 [99.8, 100.0]
Predict the mean mass (no image)	503	117.9 [109.8, 126.5]	52.0 [46.1, 57.2]

C. E3: Multi-view versus monocular on exact synthetic volumes

Scenes. We specify twenty-two Blender (Cycles) scenes with watertight food meshes whose volume is computed exactly from the mesh: eight food configurations (single rice mound, rice + dal, two idli, curry heap, lumpy rice, a three-item thali, a 40 mL small portion and a tall mound) on ceramic and polished steel plates, four textureless variants and two on banana leaf. At the time of writing, 11 had been rendered and reconstructed: rice mound, rice + dal, two idli, curry heap and lumpy rice on both plates, and the three-item thali on ceramic. Only these are reported. Each scene is rendered at 1600×1200, 60° horizontal FoV, 45° elevation and 0.40 m distance, with the plate rotating on a turntable inside a printed 50 mm ArUco mat, i.e. the rig geometry. View 0 also stores a metric depth map for a "perfect single-view depth" baseline.

Methods. (i) Dense MVS: COLMAP SfM, marker-triangulated scale, OpenMVS densification, plate-noise-calibrated footprint threshold, shared integral. The sparse-SfM, monocular and perfect-single-view baselines use the same integral and are implemented, but their runs had not finished at the time of writing and are not reported.

Footprint rule. Dense MVS leaves millimetre-level noise on the plate surface. A cell counts as food if its height exceeds $\mu + 2\sigma$ of the plate annulus heights. This rule was fixed on `rice_mound__ceramic` before the other scenes were run, and those scenes are held out. A hysteresis variant (seed at 2σ, grow connected cells down to 0.5σ) was added *after* the curry failure below, and is reported as post hoc.

Table IV gives the results. With the pre-registered footprint rule, dense MVS has a median absolute error of 3.9 % [1.0,

TABLE IV
E3 (SYNTHETIC, EXACT MESH VOLUME). ABSOLUTE VOLUME ERROR, MEDIAN AND 95% BOOTSTRAP CI OVER SCENES; BIAS IS THE MEAN SIGNED ERROR. TIME IS VM CPU SECONDS PER SCENE.

Method	Plate	n ok	—err— % median [95% CI]	bias %	$n_{\le 10\%}$	t (s)
MVS dense (24 views)	all	11/11	3.9 [1.0, 5.7]	-4.0	5	114
MVS dense (24 views)	ceramic	6/6	5.3 [1.9, 16.0]	-5.7	3	114
MVS dense (24 views)	steel	5/5	2.0 [0.2, 4.2]	-1.9	5	119
MVS dense, hysteresis (post-hoc)	all	11/11	2.0 [0.3, 6.0]	+4.4	5	117
MVS dense, hysteresis (post-hoc)	ceramic	6/6	5.5 [1.7, 21.1]	+8.3	4	114
MVS dense, hysteresis (post-hoc)	steel	5/5	0.3 [0.1, 2.0]	-0.2	5	119

5.7] over 11 scenes, and 10 of 11 are within 10 %. Steel plates (2.0 %) do better than ceramic (5.3 %). The textureless white ceramic gives plate-surface noise of 1.9–6.2 mm, against 0.9–1.2 mm on the textured steel.

Failure case. `curry_heap__ceramic` is -26.0 %: its 6.2 mm plate noise raises the threshold above the low outer skirt of the heap, which is then cut from the footprint. The post-hoc hysteresis rule recovers it (-1.0 %) and lowers the median error to 2.0 %, but its bias is $+4.4$ % and it overestimates lumpy rice on ceramic by 24.2 % and the thali by 17.9 %, where plate noise connects to the food. No single height threshold works on every scene. We keep the pre-registered rule as primary and conclude that *separating food from plate*, not depth accuracy, is now the dominant error of the multi-view pipeline. Image-space food segmentation projected onto the height field is the natural next step. Reconstruction takes about 117 s per scene on the VM CPU (VM evidence).

D. E4: Ablations

The ablations use the dense MVS pipeline and change one factor at a time: number of views (6, 8, 12, 16, 24, evenly spaced), image resolution and JPEG quality over the OV2640 range, a deliberate error in the printed marker size (± 1, ± 2 %), plate material, and three repeated runs on identical input to measure the run-to-run spread of SfM/MVS, which are not bit-deterministic. A run fails if SfM returns no model, if no marker side is triangulated, or if densification errors; failures are counted, not dropped. The driver (`experiments/synth/run_suite.py`) and the analysis script are in the repository. **Status.** The ablation runs had not finished at the time of writing, so no ablation numbers are reported here. Two observations are already available from E3. Ceramic and steel differ in plate noise by about 2–5$\times$ (plate-material ablation). Two independent reconstructions of the same eight scenes differed by up to 2.7 percentage points (e.g. `idli_pair__ceramic` -8.4 vs -5.7 %), which bounds run-to-run repeatability before the formal repeat experiment.

VII. LIMITATIONS

Nutrition5k has no volume ground truth. The dataset provides overhead RealSense depth and weighed mass, not food volume. We therefore score monocular models against the volume integrated from sensor depth over the same dish region (food plus plate rim above the support plane). This measures how well a model reproduces a depth sensor, not how well it measures food. Our attempt to separate food from plate with a second plate-floor plane fit gave a food volume that barely tracks weighed mass (Spearman $\rho = 0.24$, Pearson $r = -0.01$, $n = 502$), so we do not use it as a target. Absolute food-volume accuracy is reported only on synthetic scenes with exact mesh volume.

Synthetic-to-real gap. Blender scenes have perfect camera intrinsics, controlled lighting and procedural textures. Real OV2640 frames add rolling shutter, auto-exposure drift, lens distortion and JPEG blocking on low-texture food. The resolution and JPEG ablations bound part of this gap but not all of it.

No physical measurements yet. Load-cell linearity, repeatability and drift, turntable angular error, scan duration and Wi-Fi transfer latency have scripts and empty tables (Sec. IV); none of the numbers in this paper come from the physical rig. Timings are CPU timings on the development VM, not on the ESP32-S3.

Volume is not mass. Converting volume to grams needs a per-food density. We do not estimate density; the load cell measures mass directly, which is the reason it is part of the rig.

VIII. CONCLUSION

We built a capture rig under a ˜INR 5,000 budget and a benchmark that scores monocular and multi-view volume estimation with the same integral, explicit calibration protocols and confidence intervals. On the partial results, three findings stand. First, a single overhead image anchored only on the table does not recover food volume, whatever the depth model. Second, low depth error (AbsRel) does not imply low volume error. Third, 24-view dense MVS with marker scale reaches a median error of a few percent on exact synthetic volumes, and its remaining error comes from food–plate segmentation. Volume from depth does not predict grams on real mixed dishes, which justifies the load cell. Next we will complete the model and ablation runs, add image-space food segmentation, and run the physical bench tests and real-plate captures.

REPRODUCIBILITY

All code, scene specifications, per-dish result files and the scripts that generate every table and figure are in the repository. Evidence categories are labelled throughout: *Nutrition5k* (public real RGB-D data), *synthetic* (Blender/Cycles renders with exact mesh volume), *VM* (timings on an 8-core CPU-only cloud VM, not on the ESP32), and *physical* (pending rig measurements).

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